

Khan Academy Cover Letter

<https://boards.greenhouse.io/khanacademy/jobs/4951287>

Things to include

World needs a new way to learn

Possibilities of AI

Any way to include interest based motivation?

Hand

Gas

Wet

b.c.

Story, judge

Think about implementation options with google classroom, kahoot, gamification

Try to see what already exists

Cover Letter

I am writing to express my interest in the Senior Product Manager, AI/Teachers position at Khan Academy. My resume is enclosed and highlights how my past work aligns with the requirements of the position. The purpose of this cover letter is to provide additional insight into my qualifications and my genuine enthusiasm in the opportunity with your organization.

Example Demonstration Included!

Additionally, I have included an example demonstrating how to interact between KhanMigo and ChatGPT to communicate with users.

To begin with, I have crafted a REQUEST that specifies the desired response format, provides instructions, and offers an example for ChatGPT to follow. This allows for clear communication and sets expectations for the output. Furthermore, I have included a real response from ChatGPT that adheres to the instructions and output format, thereby demonstrating successful processing of the input request.

Moreover, I have outlined the necessary steps and logic for interacting with ChatGPT on the Khan Academy side. This entails accepting user input, constructing a request to OpenAI, and relaying the response back to the user. By providing these details, I aim to illustrate a comprehensive understanding of KhanMigo's implementation and the methods for effectively interacting with OpenAI.

It is important to note that this example serves as a demonstration of my functional understanding in applications and interacting with OpenAI. I am enthusiastic about providing additional examples and feedback for KhanMigo and Khan Academy upon your request. I am eager to engage in further discussions to tailor my offerings to align with your specific interests and requirements.

Lifelong Learning

I believe that learning is a lifelong journey, and I am always eager to learn new things. Recently my compass has guided me towards AI(OpenAI), learning, and health. In my spare time, I enjoy watching educational videos, taking online courses, reading books, exercising, and meditating. I believe that learning helps me to grow as a person and become a more well-rounded individual. Each precious fragment of knowledge is like a new spell in my arsenal, empowering me to take on even greater challenges.

I have **personal** experience with some of the challenges that learners face. Having ADHD I had a difficult time paying attention in the traditional classroom environment. As a result, my focus was drawn towards the stimulating and

challenging environment of video games. Most of my life I had not understood what it meant to have a slow processing speed or to be ADHD.

After college, I changed my focus to learning and career growth. I found that I was more passionate about learning when I could learn at my own pace and tackle real-world problems. This allowed me to research, learn, and develop solutions in a way that was both challenging and rewarding.

In recent years, I started unraveling the intricacies of my strengths and weaknesses. Navigating challenges, and harnessing my strengths, I achieved strategic mastery, empowering me to overcome obstacles and optimize my abilities.

As time passed, I gradually came to acknowledge my identity as a visual learner and embarked on a quest to discover learning strategies that resonate with my style of comprehension. Throughout this quest, I discovered that incorporating visual aids, binaural beats, exercise, proper nutrition, meditation, breaks, and organization empowers me to stay focused throughout the day and eliminate the need for medication. This transformative journey has bestowed upon me a distinctive perspective, a lens through which I can aid others in their pursuit of knowledge.

I also delved into diverse methods of communication and learning, exploring verbal and non-verbal communication, associations, and emotional involvement. My journey led me to discover some of the foundational tools for teaching and learning including colors, numbers, words, symbols, sound, interactions, repetition, breaks, and visualization. Furthermore, I recognized the importance of engagement in learning, encompassing elements such as feedback, establishing purpose and relevance, rewards, growth, having defined paths and being involved in decision making. I cannot express enough how thrilled and grateful I am for the eye-opening discoveries I have made in my exploration of teaching and learning, and I am excited to integrate these insights into my educational practices.

The Education industry appeals to a diverse range of individuals who strive to make the world a better place. It attracts people who are eager to challenge themselves, expand their knowledge, and foster personal growth. I am one of these individuals, and I want to surround myself with kindred spirits who share this drive and commitment to creating a positive impact.

The belief that education is a key to unlocking boundless potential resonates with me like a symphony of pixels and sound. It urges me to champion equal opportunities for anyone, anywhere. I would love to contribute my skills and passion towards such a meaningful endeavor.

Embracing Khan Academy's Vision

I wholeheartedly align with Sal's vision and recognize him as a genuine visionary and an invaluable catalyst for positive transformation. Education is of the utmost importance, and I firmly believe in the importance of equipping the next generation with the tools for self-paced learning, revolutionizing traditional classrooms, and incrementally improving the year-based educational model to embrace a mastery-based system.

Khanmigo, in my view, holds immense potential in bringing us closer to this desired mastery-oriented approach. As an emerging platform, Khanmigo is only beginning to unveil its true capabilities, and I am eagerly anticipating its future trajectory. Sal has truly excelled in demonstrating a commendable approach to integrating AI-based technologies rather than restricting them. The profound impact that Khanmigo can have on the realm of education by offering every student a free tutor and providing teachers with invaluable teaching assistants is an essential step toward ensuring universal student success. Khanmigo effectively breaks down barriers such as the fear of asking questions and streamlines the process of seeking assistance.

Moreover, Khanmigo has the prospective ability to adapt to fuel individual learning styles and draw insights from a student's learning history. Over time, it will develop a repository of frequently asked questions in various subject areas, enabling personalized adjustments. The future holds immense promise for Khanmigo, and its potential to transform education is undeniably bright.

AI, Logic, and Education

In college, I pursued an AI course that proved to be instrumental in shaping my understanding of artificial intelligence. As part of the course, I undertook a challenging project centered around creating a Chess AI. This involved implementing heuristics and incorporating the mini-max algorithm to enable the AI to play against other Chess AI implementations and human opponents autonomously. This experience sharpened my problem-solving skills and deepened my knowledge of AI principles.

Furthermore, I have actively applied my expertise in algorithms and programming to develop diverse applications. For instance, I have successfully designed and implemented applications that generate dynamic PDF documents for real estate leasing, manage real estate inventory, determine lead heat index, and streamline follow-up processes. Over the years, I have gained substantial experience in working with data visualization and analysis, along with a expertise in leveraging APIs and developing web applications.

Beyond my technical endeavors, I have a keen interest in exploring the fascinating realm of human AI interaction. I have been actively delving into understanding the inner workings of the human mind, studying the ways in which our cognitive processes parallel those of AI. I find it intriguing how our minds resemble AI models, with our perceptions shaped by learned weights and biases based on past experiences.

Similar to how we can prompt ChatGPT to imitate the Great Gatsby, we possess the unique ability to approach a situation from various angles and perspectives. In essence, we are intricate systems akin to computers, constantly processing information and making decisions, relying heavily on RAM, neural networks, associations, and symbols.

My passion for AI language models and their potential continues to grow. These technologies have made significant strides, exhibiting remarkable efficiency and human-like capabilities. Through my exploration of ChatGPT, I have begun to unveil a world of possibilities for information processing, engaging conversations, captivating content creation, and comprehensive research. Witnessing the incredible potential of these AI tools, I am eager to contribute to their implementation in education, an area where their impact can be truly transformative, empowering students and educators alike to reach new heights of learning and teaching excellence.

Bridging the Gap: Collaborating for Educational Advancement

The thought of working with Khan Academy and having a profound and positive impact on the lives of millions sparks a fire within me, and I am thrilled at the prospect of creating meaningful change in the world

With my combination of passion and experience for technical development, communication, and product management, I bring a highly qualified skill set to excel in this role. I eagerly anticipate the opportunity to engage in a conversation and explore how my skills align with the specific needs of your esteemed organization.

EXAMPLE IMPLEMENTATION FOR UTILIZING CHATGPT TO ASSIST WITH WRITING IN KHANMIGO

Description

This sample demonstrates how to interact with AI tools and enhance KhanMigo! It can be expanded further to include specific response formatting, more descriptive for how to provide feedback (i.e. be concise), add a scoring system for writing, and many more test cases.

Example Use Case(s):

User asks KhanMigo help revising text for an essay

Example user input: "I would like help revising text for the following essay"

OR

Before submitting an answer to Khan Academy the user is presented with an option by KhanMigo to review the text and help revise before continuing.

(etc)

Example User Input:

"The enthusiastic gardener carefully planted seeds just outside the corporate office. With patience, he awaited their sprouting, tending to them by providing regular water and care. Over time, his efforts brought forth fruit, as the trees grew into magnificent fruit-bearing trees, not only producing nourishing food but also scattering seeds to propagate new growth among their fellow trees."

[BEGIN KHANMIGO RESPONSE]:

{ one of the following: [Great, Excellent, Outstanding, Superb, Splendid] }! Here are some ways that I can help you with revising your text. Please specify one or more options for how to help you revise.

Common Actions

1. Grammar Analysis and Suggestions
2. Tone Analysis and Suggestions
3. Adding or Improving a Theme
4. Options for polishing, enhancing, refining, and revising

Advanced Actions

5. Improving Sensory Details, enhancing the imagery

6. Adding a personal touch

Please respond with a list of the options you would like for me to help you,
i.e. "1, 2"

(Alternatively introduce check boxes or some other form of user input)

[END KHANMIGO TURN]

[BEGIN ALLOWING USER INPUT]

If the users response does not match the expected format, reject the request and ask for the user to input the information again. Tell them to match the following input pattern. "1, 2, 3, 4". Repeat 3 times and if the response does not come back as expected then cancel the session.

If the user does respond as expected store this response as a variable
USER_RESPONSE_TEXT_ASSISTANCE_OPTIONS.

[END USER INPUT TURN]

[interact with chatgpt based on user input]

[LOGIC HANDLING – BUILD CHATGPT REQUEST]

BASE CHATGPT REQUEST

USER_INPUT_TEXT: "{{user's input}}"

EXAMPLE_TEXT: "The enthusiastic gardener carefully planted seeds just outside the corporate office. With patience, he awaited their sprouting, tending to them by providing regular water and care. Over time, his efforts brought forth fruit, as the trees grew into magnificent fruit-bearing trees, not only producing nourishing food but also scattering seeds to propagate new growth among their fellow trees."

I will provide an example response format that references EXAMPLE_TEXT for demonstration purposes. I want you to follow this example response format when responding. However, I want you to respond following the example format

for the input titled USER_INPUT_TEXT.

Below is the example utilizing EXAMPLE_TEXT

IF 1 IS INCLUDED (Grammar Analysis and Suggestions)

if USER_RESPONSE_TEXT_ASSISTANCE_OPTIONS includes the option 1 add the following to the base request

GRAMMAR ANALYSIS

[BASED ON THE PROVIDED INPUT, ANALYZE THE GRAMMAR OF THE TEXT AND ADD A SECTION TO YOUR RESPONSE THAT FOLLOWS THE FOLLOWING FORMAT]

[STATIC TITLE]

Words that can be easily updated to improve the theme and personality of the writing:

[PROVIDE A LIST OF UP TO 4 WORDS THAT CAN EASILY BE UPDATED TO IMPROVE THE THEME AND PERSONALITY OF THE WRITING. ONLY PROVIDE A SINGLE WORD, DO NOT PROVIDE A DESCRIPTION.]

1. Enthusiastic
2. Carefully
3. Corporate
4. Patience

[static title]

Analysis of grammar in the paragraph:

[PROVIDE UP TO 4 SUGGESTIONS FOR IMPROVING THE GRAMMAR OF THE USER INPUT]

1. The use of "just" before "outside the corporate office" could be removed to improve the flow of the sentence. (e.g., "outside the corporate office")
2. "Awaited their sprouting" can be modified to "awaited their growth" or "awaited their emergence" to add variety and avoid repetition.
3. "As the trees grew into magnificent fruit-bearing trees" can be revised to "The trees flourished into magnificent fruit-bearing ones" to avoid repetition of "trees" and provide a more dynamic description.
4. "Scattering seeds to propagate new growth among their fellow trees" can be rephrased as "Dispensing seeds to foster new growth amidst their companion trees" to vary the language and enhance the imagery.

[provide a brief summary of the grammar quality]

Overall, the grammar in the paragraph is quite solid, with just a few

suggestions for minor improvements.

IF 2 IS INCLUDED (Tone Analysis and Suggestions)

if USER_RESPONSE_TEXT_ASSISTANCE_OPTIONS includes the option 2 add the following to the base request

Tone Analysis

[BASED ON THE PROVIDED INPUT, ANALYZE THE GRAMMAR OF THE TEXT AND ADD A SECTION TO YOUR RESPONSE THAT FOLLOWS THE FOLLOWING FORMAT]

Your tone: Positive

Suggestions for improvement:

1. The phrase "carefully planted" can be changed to "joyfully planted" to emphasize the gardener's happiness and excitement while carrying out the task.
2. "With patience" can be omitted to maintain a more direct and positive tone, focusing on the gardener's anticipation rather than the need for patience.
3. "As time passed, his dedicated efforts yielded abundant results as the fellow trees grew into magnificent fruit-bearing trees" can be added to emphasize the impressive growth and fruitful nature of the trees
4. The phrase "not only producing nourishing food but also scattering seeds to propagate new growth among their fellow trees" can be revised to "generously dispersed seeds, fostering new growth and camaraderie among their neighboring trees" to highlight the generous nature of the trees in spreading seeds and promoting the growth of other trees, emphasizing a positive and supportive ecosystem.

[title, analyze the tone of the input as being either positive or negative, substitute {{TONE_TYPE}} for the word positive or negative depending on the result of your analysis]

Phrases that attribute to the {{TONE_TYPE}} tone:

[include 3 examples that contribute to the overall tone]

1. "The enthusiastic gardener" - The use of the word "enthusiastic" implies a positive attitude and eagerness towards gardening.
2. "Regular water and care" - The commitment to providing regular water and care signifies the gardener's dedication and responsible approach, adding a positive tone to the description.

3. "His efforts brought forth fruit" - This phrase underscores the gardener's hard work and determination, resulting in successful outcomes.

IF 3 IS INCLUDED (Adding or Improving a Theme)

if USER_RESPONSE_TEXT_ASSISTANCE_OPTIONS includes the option 3 add the following to the base request

// TODO

IF 4 IS INCLUDED (Options for polishing, enhancing, refining, and revising)

if USER_RESPONSE_TEXT_ASSISTANCE_OPTIONS includes the option 4 add the following to the base request

// TODO

IF 5 IS INCLUDED (Improving Sensory Details, enhancing the imagery)

if USER_RESPONSE_TEXT_ASSISTANCE_OPTIONS includes the option 5 add the following to the base request

// TODO

IF 6 IS INCLUDED (Adding a personal touch)

if USER_RESPONSE_TEXT_ASSISTANCE_OPTIONS includes the option 6 add the following to the request

// TODO

APPEND ENDING TEXT

Add "This is the end of the example. Please follow this format and repeat for USER_INPUT_TEXT" to the end of the base request.

[END CHATGPT REQUEST PREPRATION]

[SEND REQUEST TO CHATGPT]

[PRESENT RESPONSE]

EXAMPLE REQUEST

USER_INPUT_TEXT: "Free education opens doors to limitless opportunities, empowering individuals to reach their full potential and shape a brighter future. OpenAI further amplifies this empowerment, equipping learners with cutting-edge AI technology to accelerate their progress."

EXAMPLE_TEXT: "The enthusiastic gardener carefully planted seeds just outside the corporate office. With patience, he awaited their sprouting, tending to them by providing regular water and care. Over time, his efforts brought forth fruit, as the trees grew into magnificent fruit-bearing trees, not only producing nourishing food but also scattering seeds to propagate new growth among their fellow trees."

I will provide an example response format that references EXAMPLE_TEXT for demonstration purposes. I want you to follow this example response format when responding. However, I want you to respond following the example format for the input titled USER_INPUT_TEXT.

Below is the example utilizing EXAMPLE_TEXT

GRAMMAR ANALYSIS

[BASED ON THE PROVIDED INPUT, ANALYZE THE GRAMMAR OF THE TEXT AND ADD A SECTION TO YOUR RESPONSE THAT FOLLOWS THE FOLLOWING FORMAT]

[STATIC TITLE]

Words that can be easily updated to improve the theme and personality of the writing:

[PROVIDE A LIST OF UP TO 4 WORDS THAT CAN EASILY BE UPDATED TO IMPROVE THE THEME AND PERSONALITY OF THE WRITING. ONLY PROVIDE A SINGLE WORD, DO NOT PROVIDE A DESCRIPTION.]

1. Enthusiastic
2. Carefully
3. Corporate
4. Patience

[static title]

Analysis of grammar in the paragraph:

[PROVIDE UP TO 4 SUGGESTIONS FOR IMPROVING THE GRAMMAR OF THE USER INPUT]

1. The use of "just" before "outside the corporate office" could be removed to improve the flow of the sentence. (e.g., "outside the corporate office")

2. "Awaited their sprouting" can be modified to "awaited their growth" or "awaited their emergence" to add variety and avoid repetition.
3. "As the trees grew into magnificent fruit-bearing trees" can be revised to "The trees flourished into magnificent fruit-bearing ones" to avoid repetition of "trees" and provide a more dynamic description.
4. "Scattering seeds to propagate new growth among their fellow trees" can be rephrased as "Dispensing seeds to foster new growth amidst their companion trees" to vary the language and enhance the imagery.

[provide a brief summary of the grammar quality]

Overall, the grammar in the paragraph is quite solid, with just a few suggestions for minor improvements.

Tone Analysis

[BASED ON THE PROVIDED INPUT, ANALYZE THE GRAMMAR OF THE TEXT AND ADD A SECTION TO YOUR RESPONSE THAT FOLLOWS THE FOLLOWING FORMAT.]

Your tone: Positive

Suggestions for improvement:

- 1 The phrase "carefully planted" can be changed to "joyfully planted" to emphasize the gardener's happiness and excitement while carrying out the task.
2. "With patience" can be omitted to maintain a more direct and positive tone, focusing on the gardener's anticipation rather than the need for patience.
3. "As time passed, his dedicated efforts yielded abundant results as the fellow trees grew into magnificent fruit-bearing trees" can be added to emphasize the impressive growth and fruitful nature of the trees
4. The phrase "not only producing nourishing food but also scattering seeds to propagate new growth among their fellow trees" can be revised to "generously dispersed seeds, fostering new growth and camaraderie among their neighboring trees" to highlight the generous nature of the trees in spreading seeds and promoting the growth of other trees, emphasizing a positive and supportive ecosystem.

[title, analyze the tone of the input as being either positive or negative, substitute {{TONE_TYPE}} for the word positive or negative depending on the result of your analysis]

Phrases that attribute to the {{TONE_TYPE}} tone:

[include 3 examples that contribute to the overall tone]

1. "The enthusiastic gardener" - The use of the word "enthusiastic" implies a positive attitude and eagerness towards gardening.
2. "Regular water and care" - The commitment to providing regular water and care signifies the gardener's dedication and responsible approach, adding a positive tone to the description.
3. "His efforts brought forth fruit" - This phrase underscores the gardener's hard work and determination, resulting in successful outcomes.

This is the end of the example.

Please follow this format and repeat for USER_INPUT_TEXT

Example Response

GRAMMAR ANALYSIS

Words that can be easily updated to improve the theme and personality of the writing:

1. Free
2. Limitless
3. Empowering
4. Shape

Analysis of grammar in the paragraph:

1. The word "opens" can be changed to "unlocks" to add variety and create a stronger impact.
2. Instead of "to limitless opportunities," using "to boundless possibilities" could provide a more evocative and imaginative description.
3. "Empowering individuals to reach their full potential" can be rephrased as "Empowering individuals to unleash their full potential," which adds a dynamic touch.
4. "Shape a brighter future" can be modified to "Craft a brighter future" or "Forge a brighter future" to introduce more creative and active verbs.

Overall, the grammar in the paragraph is well-constructed, and these suggestions aim to enhance the theme and personality of the writing.

Tone Analysis

Your tone: Positive

Suggestions for improvement:

1. The word "Free" can be changed to "Liberating" to emphasize the sense of freedom and independence provided by education.
2. "Opens doors" can be replaced with "Paves the way" to evoke a more positive and empowering tone.
3. "Limitless opportunities" can be modified to "Endless possibilities" to convey a sense of boundless potential.

4. Instead of "shape a brighter future," using "mold a radiant future" or "craft a brilliant future" would add a touch of creativity and positivity.

Phrases that attribute to the positive tone:

1. "Free education opens doors to limitless opportunities" - The use of "opens doors" and "limitless opportunities" indicates a positive outlook and the vast potential education can offer.

2. "Empowering individuals to reach their full potential" - The phrase "empowering individuals" and the focus on reaching their full potential contribute to the positive tone, highlighting personal growth and empowerment.

3. "Accelerate their progress" - The term "accelerate" implies a positive notion of advancement and success, further reinforcing the positive tone.